Likelihood-Free Overcomplete ICA and Applications in Causal Discovery

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General Framework

Linear ICA assumes the following data generation model:

$$\mathbf{x} = \mathbf{A}\mathbf{s}, \tag{1}$$

where $\mathbf{x} \in \mathbb{R}^p, \mathbf{s} \in \mathbb{R}^d, \mathbf{A} \in \mathbb{R}^{p \times d}$

- The elements in $\mathbf{s}$ are supposed to be independent from each other.
- each follows a non-Gaussian distribution (or at most one of them is Gaussian).
- The goal of ICA is to recover both $\mathbf{A}$ and $\mathbf{s}$ from observed mixtures $\mathbf{x}$.
- In causal discovery, the main goal is to recover a constrained $\mathbf{A}$ matrix. When $d > p$, the problem is known as overcomplete ICA (OICA).

Liklihood-Based Methods of Source Separation

Whenever there are hidden confounders, causal discovery and with the linearity and non-Gaussian noise constraints, the corresponding causal discovery algorithms can be seen as extension of overcomplete independent componentanalysis (OICA).
Unlike regular ICA, in which the mixing matrix is invertible, OICA is not!

- Exisiting OICA algorithms typically assume a parametric distribution for the ICs.
- For example, if assuming each IC follows a Mixture of Gaussian(MoG) distribution, one can simply derive the likelihood for the observed data.

Paper's Contribution

$$\hat{\mathbf{x}} = \mathbf{A}[\hat{s}_1, \ldots, \hat{s}_d]^\intercal = \mathbf{A}[f_{\theta_1}(z_1), \ldots, f_{\theta_d}(z_d)]^\intercal = G_{\mathbf{A},\boldsymbol{\theta}}(\mathbf{z}), \tag{2}$$

where $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_d]^\intercal$ and $\mathbf{z} = [z_1, \ldots, z_d]^\intercal$.

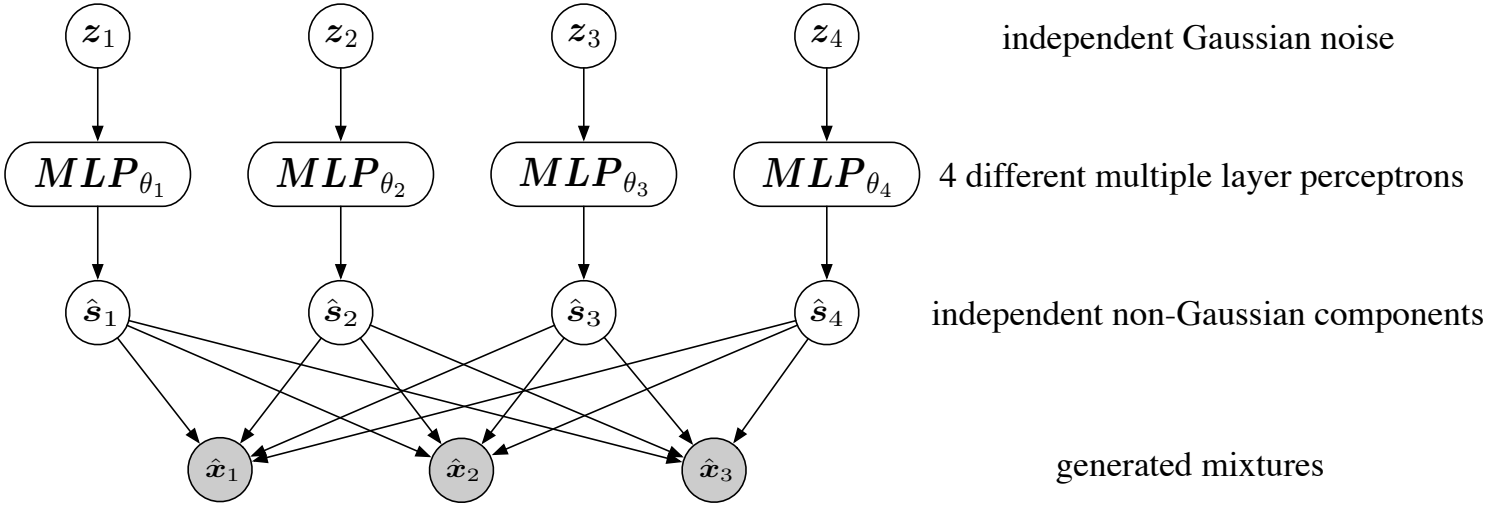

Figure: generator architecture of LFOICA. z_1, z_2, z_3, z_4 are i.i.d Gaussian noise variables.

While most of the previous algorithms for both undercomplete and overcomplete scenarios try to minimized the dependence among the recovered components, the components $\hat{s}_i$ recovered by LFOICA are essentially independent because the noises z_i are independent, according to the generating process.

The LFOICA generator $G_{\mathbf{A},\boldsymbol{\theta}}$ can be learned by minimizing the distributional distance between the data sampled from the generator and the observed $\mathbf{x}$ data.

$$\begin{aligned}\mathbf{A}^*, \boldsymbol{\theta}^* &= \underset{\mathbf{A},\boldsymbol{\theta}}{\operatorname{argmin}}\, M\left(\mathbb{P}(\mathbf{x}), \mathbb{P}(G_{\mathbf{A},\boldsymbol{\theta}}(\mathbf{z}))\right) \\ &= \underset{\mathbf{A},\boldsymbol{\theta}}{\operatorname{argmin}} \left\|\mathbb{E}_{\mathbf{x}\sim p(\mathbf{x})}[\phi(\mathbf{x})] - \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}[\phi(G_{\mathbf{A},\boldsymbol{\theta}}(\mathbf{z}))]\right\|^2, \qquad (3)\end{aligned}$$

In practice, we optimize some empirical estimator of (3) on minibatches by stochastic gradient descent (SGD).

Algorithm 1 Likelihood-Free Overcomplete ICA (LFOICA) Algorithm

1: Get a minibatch of i.i.d samples $\mathbf{z}$ from Gaussian noise distribution.
2: Generate mixtures using (2).
3: Get a minibatch of samples from the distribution of observed mixtures $p(\mathbf{x})$.
4: Update $\mathbf{A}$ and $\boldsymbol{\theta}$ by minimizing the empirical estimate of (3) on the minibatch.
5: Repeat step 1 to step 4 until max iterations reached.

Identifiability Results

Theorem

Given two OICA models $\mathbf{x} = G_{\mathbf{A},\theta}(\mathbf{x})$ *and* $\mathbf{x'} = G_{\mathbf{A}',\theta'}(\mathbf{z'})$ *that specify distributions* $\mathbb{P}(\mathbf{x})$ *and* $\mathbb{P}(\mathbf{x'})$*, respectively. Under the non-Gaussian assumption off* $f(z_i)$*, if* $MMD(\mathbb{P}(\mathbf{x'}), \mathbb{P}(\mathbf{x})) = 0$*, then* $A' = AP_pS_p$*, where* Pp *is a* $p \times p$ *column permutation matrix and* S_p *is a* $p \times p$ *scaling matrix.*

- Cites an old ICA paper (in TSP) for proof! This theorem was only added in the final version of paper.

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- θ is NOT identifiable!

Practical Consideration

- Mixing matrix is sparse in many real systems, we add a LASSO regularizer to (3), resulting in the loss function $M\left(\mathbb{P}(\mathbf{x}), \mathbb{P}(G_{\mathbf{A},\boldsymbol{\theta}}(\mathbf{z}))\right) + \lambda \sum_i \sum_j |\mathbf{A}_{ij}|$.
- Typical proximal methods can be used to solve this problem.

Causal Discovery under Measurement Error

- LiNGAM model for variables without measurement error: $\mathbf{X} = \mathbf{B}\tilde{\mathbf{X}} + \tilde{\mathbf{E}}$
- Measurement error: $\mathbf{X} = \tilde{\mathbf{X}} + \mathbf{E}$

$$\mathbf{X} = \tilde{\mathbf{X}} + \mathbf{E} = (\mathbf{I} - \mathbf{B})^{-1}\tilde{\mathbf{E}} + \mathbf{E} = \begin{bmatrix} (\mathbf{I} - \mathbf{B})^{-1} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{E}} \\ \mathbf{E} \end{bmatrix} \tag{4}$$

- Fits well within the OICA framework.

Subsampled Time Series

- Original "Causal Frequency" follow VAR(1):

$$\mathbf{x}_t = \mathbf{A}\mathbf{x}_{t-1} + \mathbf{e}_t, \tag{5}$$

- Sum-sampling with factor k:

$$\tilde{\mathbf{x}}_{t+1} = \mathbf{A}^k \tilde{\mathbf{x}}_t + \mathbf{L}\tilde{\mathbf{e}}_{t+1} \tag{6}$$

Where $\tilde{\mathbf{x}}$ is the observed data subsampled from $\mathbf{x}$, $\mathbf{L} = [\mathbf{I}, \mathbf{A}, \mathbf{A}^2, ..., \mathbf{A}^{\mathbf{k}-1}], \tilde{\mathbf{e}} = (\mathbf{e}^\top_{1+tk-0}, \mathbf{e}^\top_{1+tk-1}, ..., \mathbf{e}^\top_{1+tk-(k-1)})^\top$. We are interested in estimating the transition matrix $\mathbf{A}$ from the subsampled data.

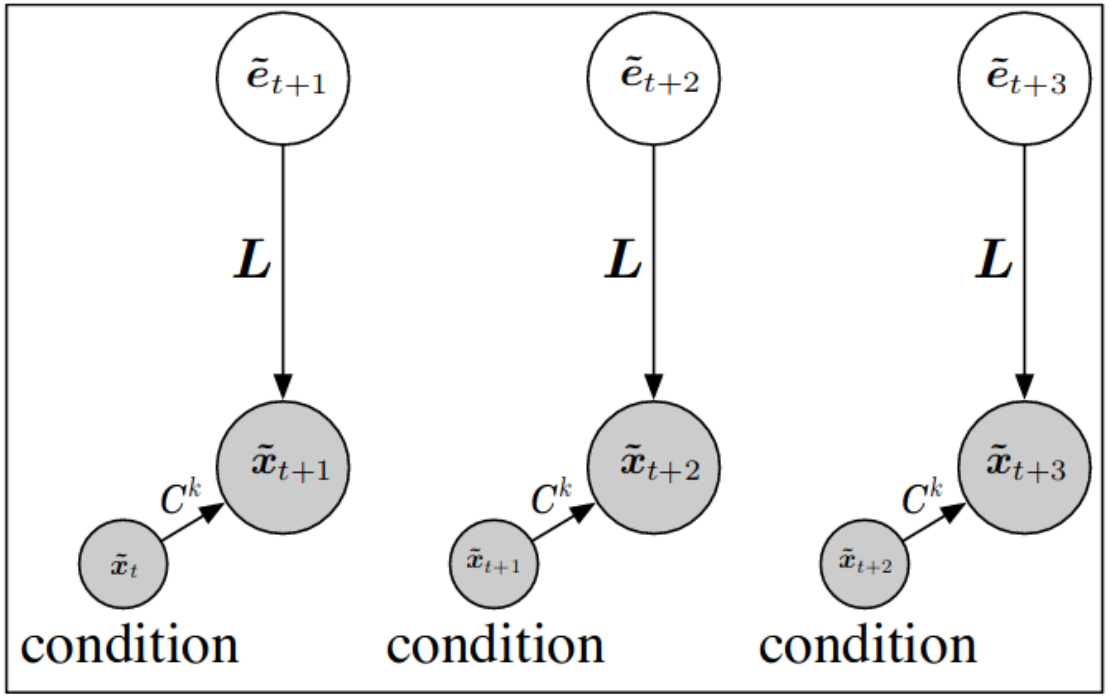
Idea! Conditional GAN

$$\tilde{\mathbf{x}}_{t+1} = \mathbf{A}^k \tilde{\mathbf{x}}_t + \mathbf{L}\tilde{\mathbf{e}}_{t+1} \tag{7}$$

Algorithm 2 LFOICA-conditional For Subsampled Data

1: Get a minibatch of i.i.d samples $\mathbf{z}$ from Gaussian noise distribution.
2: Mapping $\mathbf{z}$ using parameterized $f_{\boldsymbol{\theta}}$, get non-Gaussian noise $\hat{\tilde{\mathbf{e}}}_{t+1}$.
3: Get a minibatch of samples $\tilde{\mathbf{x}}$ from observed dataset $(\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2, ..., \tilde{\mathbf{x}}_N)$ as conditions. N is the total number of data points.
4: For each $\tilde{\mathbf{x}}_t$ in the minibatch obtained in step 3, take the corresponding data of next time step $\tilde{\mathbf{x}}_{t+1}$ from $(\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2, ..., \tilde{\mathbf{x}}_N)$ as targets.
5: Generator takes non-Gaussian noise $\hat{\tilde{\mathbf{e}}}_{t+1}$ and conditions $\tilde{\mathbf{x}}_t$ as inputs and generate target data $\hat{\tilde{\mathbf{x}}}_{t+1} = \mathbf{C}^k \tilde{\mathbf{x}}_t + \mathbf{L}\hat{\tilde{\mathbf{e}}}_{t+1}$, where $\mathbf{L} = [\mathbf{I}, \mathbf{C}, \mathbf{C}^2, ..., \mathbf{C}^{k-1}]$.
6: Update $\mathbf{C}$ and $\boldsymbol{\theta}$ by minimizing the empirical estimate of distributional distance between $p(\hat{\tilde{\mathbf{x}}}_{t+1}|\tilde{\mathbf{x}}_t)$ and $p(\tilde{\mathbf{x}}_{t+1}|\tilde{\mathbf{x}}_t)$ on the minibatch.
7: Repeat step 1 to step 6 until max iterations reached.

$$\mathbf{C}^*, \boldsymbol{\theta}^* = \arg\min_{\mathbf{C},\boldsymbol{\theta}} M\left(\mathbb{P}(\tilde{\mathbf{x}}_t, \tilde{\mathbf{x}}_{t+1}), \mathbb{P}(G_{\mathbf{C},\boldsymbol{\theta}}(\tilde{\mathbf{x}}_t, \mathbf{z}_{t+1}), \tilde{\mathbf{x}}_{t+1})\right)$$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